line 16 is necessary if y's right child is $T.nil$, and otherwise it does not change anything.

Otherwise, node z is either the same as node y or it is a proper ancestor of y's original parent. In these cases, the calls of RB-TRANSPLANT in lines 5, 8, and 13 set $x.p$ correctly in line 6 of RB-TRANSPLANT. (In these calls of RB-TRANSPLANT, the third parameter passed is the same as x.)

- Finally, if node y was black, one or more violations of the red-black properties might arise. The call of RB-DELETE-FIXUP in line 22 restores the red-black properties. If y was red, the red-black properties still hold when y is removed or moved, for the following reasons:

  1. No black-heights in the tree have changed. (See Exercise 13.4-1.)
  2. No red nodes have been made adjacent. If z has at most one child, then y and z are the same node. That node is removed, with a child taking its place. If the removed node was red, then neither its parent nor its children can also be red, so moving a child to take its place cannot cause two red nodes to become adjacent. If, on the other hand, z has two children, then y takes z's place in the tree, along with z's color, so there cannot be two adjacent red nodes at y's new position in the tree. In addition, if y was not z's right child, then y's original right child x replaces y in the tree. Since y is red, x must be black, and so replacing y by x cannot cause two red nodes to become adjacent.
  3. Because y could not have been the root if it was red, the root remains black.

If node y was black, three problems may arise, which the call of RB-DELETE-FIXUP will remedy. First, if y was the root and a red child of y became the new root, property 2 is violated. Second, if both x and its new parent are red, then a violation of property 4 occurs. Third, moving y within the tree causes any simple path that previously contained y to have one less black node. Thus, property 5 is now violated by any ancestor of y in the tree. We can correct the violation of property 5 by saying that when the black node y is removed or moved, its blackness transfers to the node x that moves into y's original position, giving x an "extra" black. That is, if we add 1 to the count of black nodes on any simple path that contains x, then under this interpretation, property 5 holds. But now another problem emerges: node x is neither red nor black, thereby violating property 1. Instead, node x is either "doubly black" or "red-and-black," and it contributes either 2 or 1, respectively, to the count of black nodes on simple paths containing x. The *color* attribute of x will still be either RED (if x is red-and-black) or BLACK (if x is doubly black). In other words, the extra black on a node is reflected in x's pointing to the node rather than in the *color* attribute.

The procedure RB-DELETE-FIXUP on the next page restores properties 1, 2, and 4. Exercises 13.4-2 and 13.4-3 ask you to show that the procedure restores properties 2 and 4, and so in the remainder of this section, we focus on property 1. The goal of the **while** loop in lines 1–43 is to move the extra black up the tree until

1. x points to a red-and-black node, in which case line 44 colors x (singly) black;
2. x points to the root, in which case the extra black simply vanishes; or
3. having performed suitable rotations and recolorings, the loop exits.

Like RB-INSERT-FIXUP, the RB-DELETE-FIXUP procedure handles two symmetric situations: lines 3–22 for when node x is a left child, and lines 24–43 for when x is a right child. Our proof focuses on the four cases shown in lines 3–22.

Within the **while** loop, x always points to a nonroot doubly black node. Line 2 determines whether x is a left child or a right child of its parent $x.p$ so that either lines 3–22 or 24–43 will execute in a given iteration. The sibling of x is always denoted by a pointer w. Since node x is doubly black, node w cannot be $T.nil$, because otherwise, the number of blacks on the simple path from $x.p$ to the (singly black) leaf w would be smaller than the number on the simple path from $x.p$ to x.

Recall that the RB-DELETE procedure always assigns to $x.p$ before calling RB-DELETE-FIXUP (either within the call of RB-TRANSPLANT in line 13 or the assignment in line 16), even when node x is the sentinel $T.nil$. That is because RB-DELETE-FIXUP references x's parent $x.p$ in several places, and this attribute must point to the node that became x's parent in RB-DELETE—even if x is $T.nil$.

Figure 13.7 demonstrates the four cases in the code when node x is a left child. (As in RB-INSERT-FIXUP, the cases in RB-DELETE-FIXUP are not mutually exclusive.) Before examining each case in detail, let's look more generally at how we can verify that the transformation in each of the cases preserves property 5. The key idea is that in each case, the transformation applied preserves the number of black nodes (including x's extra black) from (and including) the root of the subtree shown to the roots of each of the subtrees $\alpha, \beta, \ldots, \zeta$. Thus, if property 5 holds prior to the transformation, it continues to hold afterward. For example, in Figure 13.7(a), which illustrates case 1, the number of black nodes from the root to the root of either subtree α or β is 3, both before and after the transformation. (Again, remember that node x adds an extra black.) Similarly, the number of black nodes from the root to the root of any of γ, δ, ε, and ζ is 2, both before and after the transformation.[2] In Figure 13.7(b), the counting must involve the value c of the *color* attribute of the root of the subtree shown, which can be either RED or BLACK.

---

[2] If property 5 holds, we can assume that paths from the roots of $\gamma, \delta, \varepsilon$, and ζ down to leaves contain one more black than do paths from the roots of α and β down to leaves.

```
RB-DELETE-FIXUP(T, x)
 1  while x ≠ T.root and x.color == BLACK
 2      if x == x.p.left                    // is x a left child?
 3          w = x.p.right                   // w is x's sibling
 4          if w.color == RED
 5              w.color = BLACK
 6              x.p.color = RED
 7              LEFT-ROTATE(T, x.p)                          } case 1
 8              w = x.p.right
 9          if w.left.color == BLACK and w.right.color == BLACK
10              w.color = RED                                } case 2
11              x = x.p
12          else
13              if w.right.color == BLACK
14                  w.left.color = BLACK
15                  w.color = RED
16                  RIGHT-ROTATE(T, w)                       } case 3
17                  w = x.p.right
18              w.color = x.p.color
19              x.p.color = BLACK
20              w.right.color = BLACK                        } case 4
21              LEFT-ROTATE(T, x.p)
22              x = T.root
23      else // same as lines 3–22, but with "right" and "left" exchanged
24          w = x.p.left
25          if w.color == RED
26              w.color = BLACK
27              x.p.color = RED
28              RIGHT-ROTATE(T, x.p)
29              w = x.p.left
30          if w.right.color == BLACK and w.left.color == BLACK
31              w.color = RED
32              x = x.p
33          else
34              if w.left.color == BLACK
35                  w.right.color = BLACK
36                  w.color = RED
37                  LEFT-ROTATE(T, w)
38                  w = x.p.left
39              w.color = x.p.color
40              x.p.color = BLACK
41              w.left.color = BLACK
42              RIGHT-ROTATE(T, x.p)
43              x = T.root
44  x.color = BLACK
```

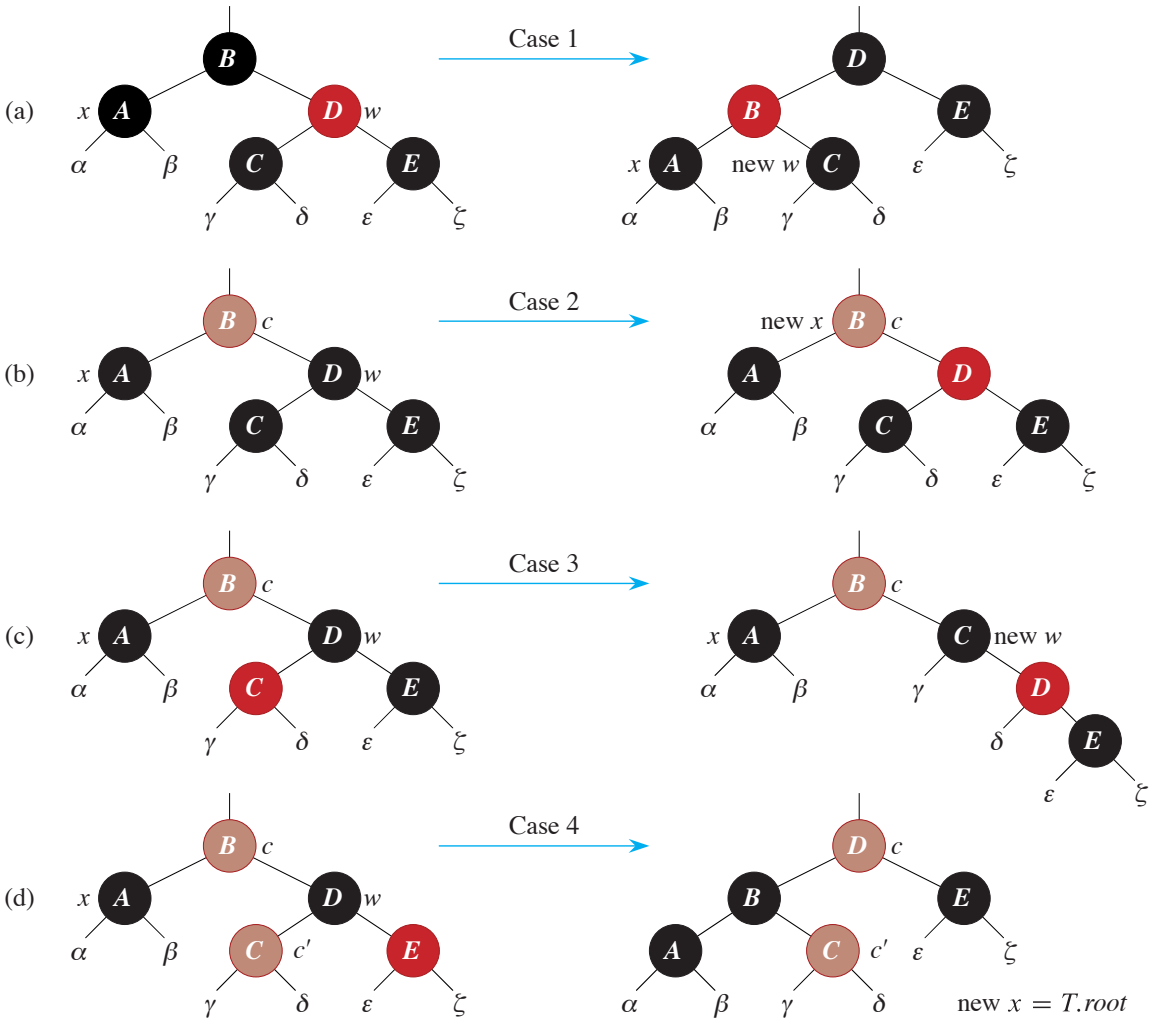


Figure 13.7 The cases in lines 3–22 of the procedure RB-DELETE-FIXUP. Brown nodes have *color* attributes represented by c and c', which may be either RED or BLACK. The letters $\alpha, \beta, \ldots, \zeta$ represent arbitrary subtrees. Each case transforms the configuration on the left into the configuration on the right by changing some colors and/or performing a rotation. Any node pointed to by x has an extra black and is either doubly black or red-and-black. Only case 2 causes the loop to repeat. **(a)** Case 1 is transformed into case 2, 3, or 4 by exchanging the colors of nodes B and D and performing a left rotation. **(b)** In case 2, the extra black represented by the pointer x moves up the tree by coloring node D red and setting x to point to node B. If case 2 is entered through case 1, the **while** loop terminates because the new node x is red-and-black, and therefore the value c of its *color* attribute is RED. **(c)** Case 3 is transformed to case 4 by exchanging the colors of nodes C and D and performing a right rotation. **(d)** Case 4 removes the extra black represented by x by changing some colors and performing a left rotation (without violating the red-black properties), and then the loop terminates.

If we define count(RED) $= 0$ and count(BLACK) $= 1$, then the number of black nodes from the root to α is $2 +$ count(c), both before and after the transformation. In this case, after the transformation, the new node x has *color* attribute c, but this node is really either red-and-black (if $c =$ RED) or doubly black (if $c =$ BLACK). You can verify the other cases similarly (see Exercise 13.4-6).

Case 1: x's sibling w is red

Case 1 (lines 5–8 and Figure 13.7(a)) occurs when node w, the sibling of node x, is red. Because w is red, it must have black children. This case switches the colors of w and $x.p$ and then performs a left-rotation on $x.p$ without violating any of the red-black properties. The new sibling of x, which is one of w's children prior to the rotation, is now black, and thus case 1 converts into one of cases 2, 3, or 4.

Cases 2, 3, and 4 occur when node w is black and are distinguished by the colors of w's children.

Case 2: x's sibling w is black, and both of w's children are black

In case 2 (lines 10–11 and Figure 13.7(b)), both of w's children are black. Since w is also black, this case removes one black from both x and w, leaving x with only one black and leaving w red. To compensate for x and w each losing one black, x's parent $x.p$ can take on an extra black. Line 11 does so by moving x up one level, so that the **while** loop repeats with $x.p$ as the new node x. If case 2 enters through case 1, the new node x is red-and-black, since the original $x.p$ was red. Hence, the value c of the *color* attribute of the new node x is RED, and the loop terminates when it tests the loop condition. Line 44 then colors the new node x (singly) black.

Case 3: x's sibling w is black, w's left child is red, and w's right child is black

Case 3 (lines 14–17 and Figure 13.7(c)) occurs when w is black, its left child is red, and its right child is black. This case switches the colors of w and its left child $w.left$ and then performs a right rotation on w without violating any of the red-black properties. The new sibling w of x is now a black node with a red right child, and thus case 3 falls through into case 4.

Case 4: x's sibling w is black, and w's right child is red

Case 4 (lines 18–22 and Figure 13.7(d)) occurs when node x's sibling w is black and w's right child is red. Some color changes and a left rotation on $x.p$ allow the extra black on x to vanish, making it singly black, without violating any of the red-black properties. Line 22 sets x to be the root, and the **while** loop terminates when it next tests the loop condition.

Analysis

What is the running time of RB-DELETE? Since the height of a red-black tree of n nodes is $O(\lg n)$, the total cost of the procedure without the call to RB-DELETE-FIXUP takes $O(\lg n)$ time. Within RB-DELETE-FIXUP, each of cases 1, 3, and 4 lead to termination after performing a constant number of color changes and at most three rotations. Case 2 is the only case in which the **while** loop can be repeated, and then the pointer x moves up the tree at most $O(\lg n)$ times, performing no rotations. Thus, the procedure RB-DELETE-FIXUP takes $O(\lg n)$ time and performs at most three rotations, and the overall time for RB-DELETE is therefore also $O(\lg n)$.

Exercises

13.4-1
Show that if node y in RB-DELETE is red, then no black-heights change.

13.4-2
Argue that after RB-DELETE-FIXUP executes, the root of the tree must be black.

13.4-3
Argue that if in RB-DELETE both x and $x.p$ are red, then property 4 is restored by the call to RB-DELETE-FIXUP(T, x).

13.4-4
In Exercise 13.3-2 on page 346, you found the red-black tree that results from successively inserting the keys $41, 38, 31, 12, 19, 8$ into an initially empty tree. Now show the red-black trees that result from the successive deletion of the keys in the order $8, 12, 19, 31, 38, 41$.

13.4-5
Which lines of the code for RB-DELETE-FIXUP might examine or modify the sentinel $T.nil$?

13.4-6
In each of the cases of Figure 13.7, give the count of black nodes from the root of the subtree shown to the roots of each of the subtrees $\alpha, \beta, \ldots, \zeta$, and verify that each count remains the same after the transformation. When a node has a *color* attribute c or c', use the notation count(c) or count(c') symbolically in your count.

13.4-7
Professors Skelton and Baron worry that at the start of case 1 of RB-DELETE-FIXUP, the node $x.p$ might not be black. If $x.p$ is not black, then lines 5–6 are

wrong. Show that $x.p$ must be black at the start of case 1, so that the professors need not be concerned.

13.4-8

A node x is inserted into a red-black tree with RB-INSERT and then is immediately deleted with RB-DELETE. Is the resulting red-black tree always the same as the initial red-black tree? Justify your answer.

★ ***13.4-9***

Consider the operation RB-ENUMERATE(T, r, a, b), which outputs all the keys k such that $a \le k \le b$ in a subtree rooted at node r in an n-node red-black tree T. Describe how to implement RB-ENUMERATE in $\Theta(m + \lg n)$ time, where m is the number of keys that are output. Assume that the keys in T are unique and that the values a and b appear as keys in T. How does your solution change if a and b might not appear in T?

Problems

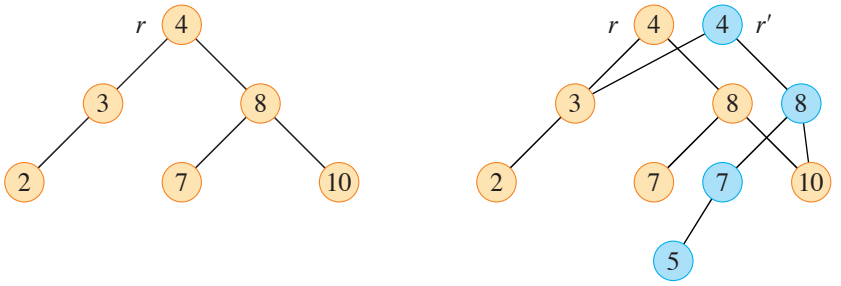
13-1 Persistent dynamic sets

During the course of an algorithm, you sometimes find that you need to maintain past versions of a dynamic set as it is updated. We call such a set ***persistent***. One way to implement a persistent set is to copy the entire set whenever it is modified, but this approach can slow down a program and also consume a lot of space. Sometimes, you can do much better.

Consider a persistent set S with the operations INSERT, DELETE, and SEARCH, which you implement using binary search trees as shown in Figure 13.8(a). Maintain a separate root for every version of the set. In order to insert the key 5 into the set, create a new node with key 5. This node becomes the left child of a new node with key 7, since you cannot modify the existing node with key 7. Similarly, the new node with key 7 becomes the left child of a new node with key 8 whose right child is the existing node with key 10. The new node with key 8 becomes, in turn, the right child of a new root r' with key 4 whose left child is the existing node with key 3. Thus, you copy only part of the tree and share some of the nodes with the original tree, as shown in Figure 13.8(b).

Assume that each tree node has the attributes *key*, *left*, and *right* but no parent. (See also Exercise 13.3-6 on page 346.)

a. For a persistent binary search tree (not a red-black tree, just a binary search tree), identify the nodes that need to change to insert or delete a node.



(a) (b)

Figure 13.8 **(a)** A binary search tree with keys $2, 3, 4, 7, 8, 10$. **(b)** The persistent binary search tree that results from the insertion of key 5. The most recent version of the set consists of the nodes reachable from the root r', and the previous version consists of the nodes reachable from r. Blue nodes are added when key 5 is inserted.

b. Write a procedure PERSISTENT-TREE-INSERT(T, z) that, given a persistent binary search tree T and a node z to insert, returns a new persistent tree T' that is the result of inserting z into T. Assume that you have a procedure COPY-NODE(x) that makes a copy of node x, including all of its attributes.

c. If the height of the persistent binary search tree T is h, what are the time and space requirements of your implementation of PERSISTENT-TREE-INSERT? (The space requirement is proportional to the number of nodes that are copied.)

d. Suppose that you include the parent attribute in each node. In this case, the PERSISTENT-TREE-INSERT procedure needs to perform additional copying. Prove that PERSISTENT-TREE-INSERT then requires $\Omega(n)$ time and space, where n is the number of nodes in the tree.

e. Show how to use red-black trees to guarantee that the worst-case running time and space are $O(\lg n)$ per insertion or deletion. You may assume that all keys are distinct.

13-2 Join operation on red-black trees

The ***join*** operation takes two dynamic sets S_1 and S_2 and an element x such that for any $x_1 \in S_1$ and $x_2 \in S_2$, we have $x_1.key \le x.key \le x_2.key$. It returns a set $S = S_1 \cup \{x\} \cup S_2$. In this problem, we investigate how to implement the join operation on red-black trees.

a. Suppose that you store the black-height of a red-black tree T as the new attribute $T.bh$. Argue that RB-INSERT and RB-DELETE can maintain the bh

attribute without requiring extra storage in the nodes of the tree and without increasing the asymptotic running times. Show how to determine the black-height of each node visited while descending through T, using $O(1)$ time per node visited.

Let T_1 and T_2 be red-black trees and x be a key value such that for any nodes x_1 in T_1 and x_2 in T_2, we have $x_1.key \leq x.key \leq x_2.key$. You will show how to implement the operation RB-JOIN(T_1, x, T_2), which destroys T_1 and T_2 and returns a red-black tree $T = T_1 \cup \{x\} \cup T_2$. Let n be the total number of nodes in T_1 and T_2.

b. Assume that $T_1.bh \geq T_2.bh$. Describe an $O(\lg n)$-time algorithm that finds a black node y in T_1 with the largest key from among those nodes whose black-height is $T_2.bh$.

c. Let T_y be the subtree rooted at y. Describe how $T_y \cup \{x\} \cup T_2$ can replace T_y in $O(1)$ time without destroying the binary-search-tree property.

d. What color should you make x so that red-black properties 1, 3, and 5 are maintained? Describe how to enforce properties 2 and 4 in $O(\lg n)$ time.

e. Argue that no generality is lost by making the assumption in part (b). Describe the symmetric situation that arises when $T_1.bh \leq T_2.bh$.

f. Argue that the running time of RB-JOIN is $O(\lg n)$.

13-3 AVL trees

An ***AVL tree*** is a binary search tree that is ***height balanced***: for each node x, the heights of the left and right subtrees of x differ by at most 1. To implement an AVL tree, maintain an extra attribute h in each node such that $x.h$ is the height of node x. As for any other binary search tree T, assume that $T.root$ points to the root node.

a. Prove that an AVL tree with n nodes has height $O(\lg n)$. (*Hint:* Prove that an AVL tree of height h has at least F_h nodes, where F_h is the hth Fibonacci number.)

b. To insert into an AVL tree, first place a node into the appropriate place in binary search tree order. Afterward, the tree might no longer be height balanced. Specifically, the heights of the left and right children of some node might differ by 2. Describe a procedure BALANCE(x), which takes a subtree rooted at x whose left and right children are height balanced and have heights that differ

by at most 2, so that $|x.right.h - x.left.h| \leq 2$, and alters the subtree rooted at x to be height balanced. The procedure should return a pointer to the node that is the root of the subtree after alterations occur. (*Hint:* Use rotations.)

c. Using part (b), describe a recursive procedure AVL-INSERT(T, z) that takes an AVL tree T and a newly created node z (whose key has already been filled in), and adds z into T, maintaining the property that T is an AVL tree. As in TREE-INSERT from Section 12.3, assume that $z.key$ has already been filled in and that $z.left = \text{NIL}$ and $z.right = \text{NIL}$. Assume as well that $z.h = 0$.

d. Show that AVL-INSERT, run on an n-node AVL tree, takes $O(\lg n)$ time and performs $O(\lg n)$ rotations.

Chapter notes

The idea of balancing a search tree is due to Adel'son-Vel'skiĭ and Landis [2], who introduced a class of balanced search trees called "AVL trees" in 1962, described in Problem 13-3. Another class of search trees, called "2-3 trees," was introduced by J. E. Hopcroft (unpublished) in 1970. A 2-3 tree maintains balance by manipulating the degrees of nodes in the tree, where each node has either two or three children. Chapter 18 covers a generalization of 2-3 trees introduced by Bayer and McCreight [39], called "B-trees."

Red-black trees were invented by Bayer [38] under the name "symmetric binary B-trees." Guibas and Sedgewick [202] studied their properties at length and introduced the red/black color convention. Andersson [16] gives a simpler-to-code variant of red-black trees. Weiss [451] calls this variant AA-trees. An AA-tree is similar to a red-black tree except that left children can never be red.

Sedgewick and Wayne [402] present red-black trees as a modified version of 2-3 trees in which a node with three children is split into two nodes with two children each. One of these nodes becomes the left child of the other, and only left children can be red. They call this structure a "left-leaning red-black binary search tree." Although the code for left-leaning red-black binary search trees is more concise than the red-black tree pseudocode in this chapter, operations on left-leaning red-black binary search trees do not limit the number of rotations per operation to a constant. This distinction will matter in Chapter 17.

Treaps, a hybrid of binary search trees and heaps, were proposed by Seidel and Aragon [404]. They are the default implementation of a dictionary in LEDA [324], which is a well-implemented collection of data structures and algorithms.

There are many other variations on balanced binary trees, including weight-balanced trees [344], k-neighbor trees [318], and scapegoat trees [174]. Perhaps